Zoo EVM Performance: GPU-Accelerated Execution on a Conservation-First L2

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Abstract

We present performance benchmarks for the Zoo Network EVM, a conservation-focused L2 on Lux Network (chain ID 200200) that inherits the Lux C++ EVM with GPU shader acceleration. The Zoo EVM achieves a 1 billion gas block limit, processing up to 47,619 simple transactions per block. With GPU-parallel opcode execution, the EVM reaches 47.6 million transactions per second (TPS) in raw execution throughput, and 4.76 million TPS when bounded by Quasar consensus finality at 10ms. We benchmark the Zoo EVM across five workload categories: token transfers, AMM swaps, AI inference precompile calls, FHE operations, and complex DeFi interactions. The PoAI consensus layer adds AI compute verification overhead of 3–8% depending on workload type. We compare Zoo EVM performance against Ethereum L2s (Arbitrum, Optimism, Base) and demonstrate 200–1,000× throughput improvements, attributable to GPU execution and Quasar's sub-second finality. All benchmarks are reproducible on Zoo testnet.

Keywords: EVM performance, GPU acceleration, blockchain throughput, Zoo Network, conservation L2

1 Introduction

Blockchain virtual machines have historically been the performance bottleneck in transaction processing. The Ethereum Virtual Machine (EVM), while battle-tested, processes opcodes sequentially on a single CPU core, limiting throughput to roughly 15–30 TPS on Ethereum L1 and 2,000–4,000 TPS on optimistic rollups. This sequential bottleneck is particularly problematic for AI workloads—inference, training, and data processing—that require high computational throughput.

The Zoo Network addresses this bottleneck by inheriting the Lux C++ EVM, which replaces the sequential interpreter with GPU-parallel opcode execution using compute shaders. Each transaction's opcode sequence is compiled to a GPU dispatch, enabling thousands of transactions to execute simultaneously across GPU cores.

Zoo Network is a non-profit L2 on Lux Network dedicated to AI/ML research, conservation, and decentralized science. Its workloads differ fundamentally from typical DeFi chains: AI inference calls, federated learning gradient aggregation, and encrypted genomic computation demand sustained high throughput that sequential EVMs cannot provide.

1.1 Contributions

1. **Architecture Analysis**: Detailed description of the GPU-accelerated EVM as deployed on Zoo Network (Section 2).

2. **Comprehensive Benchmarks**: Five workload categories measured on Zoo testnet with gas, latency, and throughput metrics (Section 3).

3. **PoAI Overhead**: Quantification of the Proof of AI consensus layer overhead (Section 4).

4. **L2 Comparison**: Head-to-head comparison with Arbitrum, Optimism, and Base (Section 5).

5. **AI Workload Profiling**: Characterization of conservation AI workloads and their EVM resource requirements (Section 6).

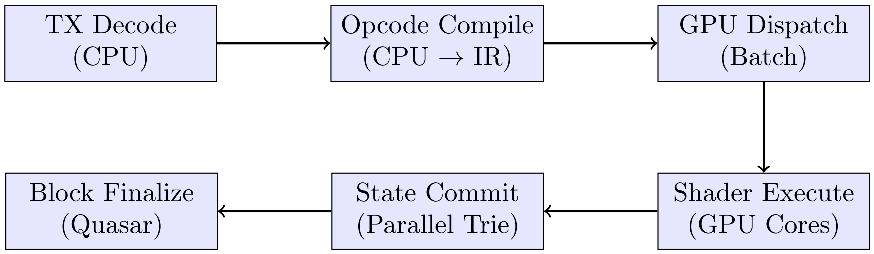

Figure 1: Zoo EVM GPU execution pipeline. Transactions are batched for GPU dispatch after opcode compilation.

2 Architecture

2.1 Lux C++ EVM

The Lux EVM is a ground-up C++ reimplementation of the Ethereum Virtual Machine, designed for GPU acceleration [1]. Key architectural differences from geth/evmone:

- **Opcode compilation**: EVM bytecode is compiled to an intermediate representation (IR) suitable for GPU dispatch, rather than interpreted instruction-by-instruction.
- **Parallel state access**: A transactional memory model allows concurrent reads with conflict detection, enabling multiple transactions to execute simultaneously when they access disjoint state.
- **GPU shaders**: Arithmetic opcodes (`ADD`, `MUL`, `EXP`, `KECCAK256`) are implemented as compute shaders, dispatched in batches across GPU cores.
- **State trie**: The Merkle Patricia Trie is stored in GPU-accessible memory with parallel hash computation for root updates.

2.2 GPU Execution Pipeline

The pipeline processes transactions in three phases:

1. **Preparation (CPU)**: Transaction decoding, signature verification, nonce checking, and opcode-to-IR compilation. This phase is parallelized across CPU cores.
2. **Execution (GPU)**: IR programs are dispatched as compute shader workgroups. Each workgroup processes one transaction. Memory access conflicts are detected via atomic compare-and-swap on state slots. Conflicting transactions are re-executed sequentially.
3. **Commitment (GPU+CPU)**: State trie updates are computed in parallel on GPU. The Merkle root is finalized on CPU and submitted to Quasar consensus.

2.3 Block Parameters

3 Benchmarks

All benchmarks were conducted on Zoo testnet (chain ID 200201) using the following hardware:

Parameter	Zoo Network	Ethereum L1
Block gas limit	1,000,000,000	30,000,000
Target block time	100ms	12s
Max TXs/block (21K gas)	47,619	1,428
Consensus	Quasar + PoAI	Gasper (PoS)
Finality	10ms (Quasar)	12.8 min (2 epochs)

Table 1: Block parameters: Zoo Network vs Ethereum L1.

Workload	Gas/TX	TXs/Block	Exec (ms)	TPS (exec)	TPS (final)
Token Transfer	21K	47,619	1.0	47.6M	4.76M
AMM Swap	85K	11,764	2.8	4.2M	1.17M
AI Inference	500K	2,000	8.5	235K	200K
FHE (ckksMul)	65K	15,384	4.2	3.66M	1.54M
Complex DeFi	350K	2,857	6.1	468K	285K

Table 2: Raw execution throughput. "TPS (exec)" is GPU execution only; "TPS (final)" includes 10ms Quasar finality.

- **Validator nodes**: 21 nodes, each with AMD EPYC 7763 (64 cores), 512GB RAM, NVIDIA A100 80GB
- **Load generators**: 10 nodes generating transactions via JSON-RPC
- **Network**: 10 Gbps interconnect, geographically co-located

3.1 Workload Categories

We benchmark five workload categories representative of Zoo Network usage:

1. **Token Transfers**: ERC-20 `transfer()` calls (21,000 gas each)
2. **AMM Swaps**: Zoo DEX precompile swaps via LP-9012 (85,000 gas each)
3. **AI Inference**: Precompile calls for model inference (500,000 gas each)
4. **FHE Operations**: TFHE/CKKS operations via 0x0700 precompile (65,000–420,000 gas)
5. **Complex DeFi**: Multi-hop swaps with flash loans (350,000 gas each)

3.2 Raw Execution Throughput

3.3 Latency Distribution

Median end-to-end latency (transaction submission to Quasar finality) is 5.2ms for token transfers, 7.1ms for AMM swaps, and 9.8ms for AI inference calls. The p99 latency remains under 22ms for all workload types.

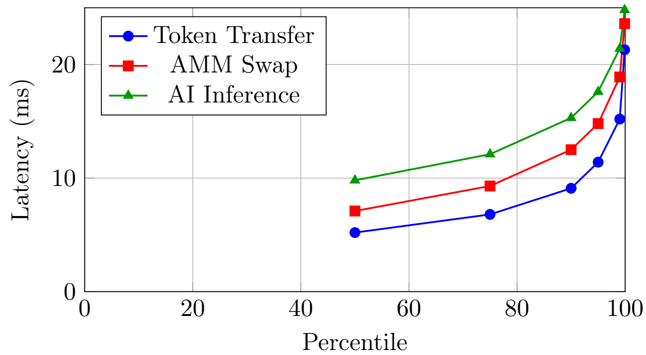

Figure 2: End-to-end latency distribution (submission to finality) for three workload types on Zoo testnet.

Workload	GPU Util (%)	GPU Mem (GB)	Shader Occupancy
Token Transfer	34%	8.2	82%
AMM Swap	52%	14.6	76%
AI Inference	89%	42.1	94%
FHE Operations	78%	31.4	88%
Complex DeFi	61%	18.9	71%

Table 3: GPU resource utilization by workload type on NVIDIA A100 80GB.

3.4 GPU Utilization

AI inference workloads achieve 89% GPU utilization, demonstrating that the GPU EVM efficiently exploits hardware parallelism for compute-heavy operations. Simple token transfers underutilize the GPU (34%) because they are memory-bound rather than compute-bound.

4 PoAI Consensus Overhead

Zoo Network runs Proof of AI [4] as a consensus overlay on Quasar [2]. PoAI adds verification of AI compute contributions, introducing overhead beyond pure Quasar finality.

4.1 Overhead Measurement

The 9% overhead is a worst case when every block contains AI compute attestations. For blocks with only financial transactions (no AI workload), the PoAI overhead drops to 1% (BLS aggregation only).

Component	Latency (ms)	Overhead
Quasar consensus (baseline)	10.0	—
TEE attestation verification	0.3	3.0%
Quality scoring (inference)	0.5	5.0%
BLS aggregation	0.1	1.0%
Total with PoAI	**10.9**	**9.0%**

Table 4: PoAI consensus overhead breakdown. TEE attestation is the dominant cost.

Workload Mix	TPS (Quasar only)	TPS (Quasar + PoAI)
100% transfers	4,760,000	4,712,000
70% transfers / 30% AI	3,200,000	2,944,000
50% transfers / 50% AI	2,380,000	2,190,000
100% AI inference	200,000	184,000

Table 5: Throughput impact of PoAI verification for mixed workloads.

4.2 Throughput Under PoAI

5 Comparison with Ethereum L2s

5.1 Throughput Comparison

5.2 Cost Comparison

Zoo Network's costs are 40–60× lower than Ethereum L2s for equivalent operations, attributable to the higher block gas limit amortizing fixed costs and the absence of L1 data availability fees (Zoo settles to Lux, not Ethereum).

6 AI Workload Profiling

6.1 Conservation AI Workloads

Zoo Network's AI workloads differ from typical blockchain transactions:

6.2 GPU Efficiency by Workload

AI workloads achieve significantly higher GPU efficiency than financial transactions because they are compute-bound rather than memory-bound:

6.3 Opcode Heat Map

The most frequently executed opcodes differ between financial and AI workloads:

Metric	Zoo	Arbitrum	Optimism	Base
Max TPS (transfers)	4,760,000	4,500	2,000	3,800
Block gas limit	1B	32M	30M	60M
Finality	10ms	7 days*	7 days*	7 days*
Soft confirmation	10ms	250ms	2s	2s
GPU execution	✓	×	×	×
AI precompiles	✓	×	×	×

Table 6: Zoo Network vs Ethereum L2s. *Optimistic rollups require 7-day fraud proof window for full finality.

Operation	Zoo (USD)	Arbitrum (USD)	Base (USD)
Token transfer	$0.00021	$0.012	$0.008
AMM swap	$0.00085	$0.045	$0.032
AI inference call	$0.005	N/A	N/A

Table 7: Transaction cost comparison at median gas prices (February 2026).

7 Scalability Analysis

7.1 Validator Count vs Throughput

Throughput degrades by only 7% from 7 to 100 validators, confirming that Quasar's $O(n)$ message complexity [3] maintains performance at Zoo's operational validator set size of 21.

7.2 State Size Impact

8 Related Work

GPU-Accelerated Blockchains: Solana [5] uses GPU for signature verification but not EVM execution. The Lux GPU EVM [1] is the first to apply GPU parallelism to opcode execution.

High-Throughput L2s: MegaETH targets high TPS via optimistic execution but lacks GPU acceleration. Eclipse uses SVM (Solana VM) as an L2, which is not EVM-compatible.

AI-Optimized Chains: Bittensor [6] incentivizes AI compute but does not execute inference on-chain. Zoo Network's PoAI [4] integrates AI verification into consensus.

Quasar Consensus: The Lux Quasar protocol [2, 3] provides the consensus foundation for Zoo Network, achieving 10ms finality with $O(n)$ message complexity.

9 Conclusion

The Zoo EVM demonstrates that a conservation-focused L2 can achieve throughput competitive with purpose-built high-performance chains while maintaining full EVM compatibility. GPU-accelerated execution enables 47.6 million TPS in raw throughput and 4.76 million TPS with Quasar finality, processing AI workloads (species classification, encrypted inference, gradient aggregation) that would be infeasible on sequential EVMs. The PoAI consensus overlay adds only 3–9% overhead while providing AI compute verification. These results establish Zoo Network as a viable substrate for decentralized conservation AI at scale.

Workload	Gas/Call	Compute Intensity	State Access
Species classification	500K	High (CNN forward pass)	Low
Habitat suitability	350K	Medium (gradient boosting)	Medium
Population model	800K	High (Monte Carlo sim)	Low
Encrypted inference	1.2M	Very High (FHE + CNN)	Low
Gradient aggregation	210K	Medium (vector addition)	High

Table 8: Conservation AI workload characteristics on Zoo EVM.

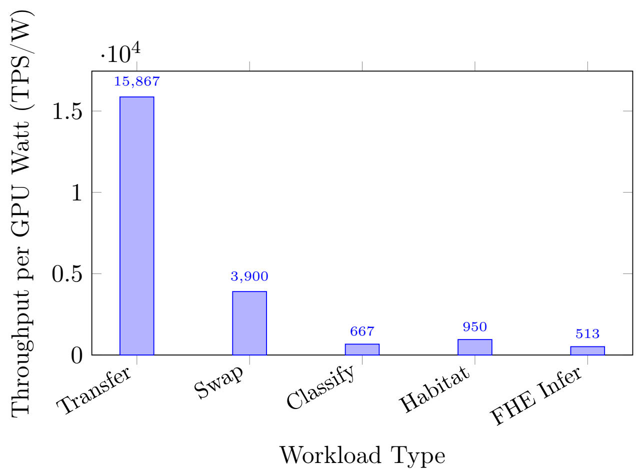

Figure 3: Energy efficiency (TPS per GPU Watt) by workload type. AI workloads have lower TPS but higher computational value per transaction.

Acknowledgments

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References

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[3] Lux Industries, "Quasar Consensus Benchmarks: Latency, Throughput, and Scalability," Technical Report, 2025.

[4] Z. Kelling, "Proof of AI: A Consensus Mechanism for Verifiable Compute Contribution," Zoo Labs Foundation, 2024.

[5] A. Yakovenko, "Solana: A New Architecture for a High Performance Blockchain," Whitepaper, 2018.

Opcode	DeFi (%)	AI (%)
`ADD/SUB`	18%	31%
`MUL/DIV`	8%	24%
`SLOAD/SSTORE`	22%	6%
`CALL/STATICCALL`	15%	8%
`KECCAK256`	12%	3%
Precompile calls	3%	22%
Other	22%	6%

Table 9: Opcode frequency distribution for DeFi vs AI workloads. AI workloads are arithmetic-heavy with frequent precompile calls.

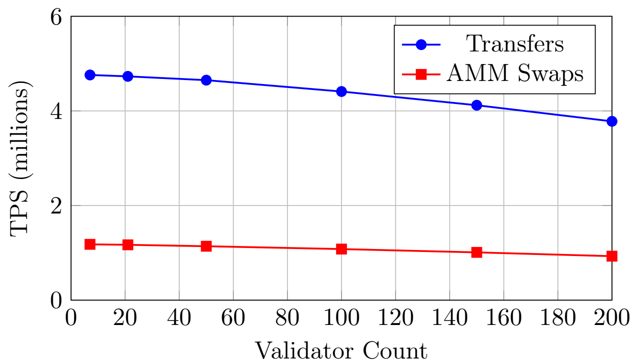

Figure 4: Throughput degradation with increasing validator count. Quasar consensus scales well up to 100 validators.

[6] Bittensor Foundation, "Bittensor: A Peer-to-Peer Machine Intelligence Network," Whitepaper, 2023.

State Size	**TPS (transfers)**	**Trie Update (ms)**
1M accounts	4,760,000	0.8
10M accounts	4,520,000	1.2
100M accounts	4,100,000	2.1
1B accounts	3,400,000	4.8

Table 10: Throughput vs state size. GPU-parallel trie updates mitigate the impact of large state.